

## CO3\_AT3 MOCK INTERVIEW

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## QUESTIONS

Q1  
Find  
10 marksQ2  
Virtual  
10 marksQ3  
Compile  
10 marksQ4  
Execute  
10 marksQ5  
Replace  
10 marksQ6  
Correct Answer  
10 marks

Q9 Bug

10 marks

Find the bug and predict the behavior. class Base { public: ~Base() { cout << "Base Destructor"; } }; class Derived : public Base { int\* data; public: Derived() { data = new int[100]; } ~Derived() { delete[] data; } }; int main() { Base\* p = new Derived(); delete p; }

## Your Answer

Bug Analysis and Behavior Prediction

1. The Bug

Non-Virtual Base Destructor: Base::~~Base() is not declared as virtual.

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Q8 Destructor

10 marks

Determine the exact destruction order. class Employee { public: Employee() {} ~Employee() { cout << "Employee "; } }; class Manager { Employee e1; Employee e2; public: ~Manager() { cout << "Manager "; } };

## Your Answer

Exact Destruction Order

1. Exact Destruction Sequence

Manager body destructor executes first, followed by member variable e2 destructor, and finally member variable e1 destructor.

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Correct Answer  
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## Q7 Operator

10 marks

Which increment operation does this represent, and what would you change for postfix counter++?

## Your Answer

Operator Overloading Analysis

1. Represented Operation

The declaration Counter& operator++() represents the Prefix Increment operation (++counter).

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Correct Answer  
10 marks

## Q6 Correct Answer

10 marks

Choose the correct smart pointer: A Company owns its Employee objects and employees should be destroyed when the company is destroyed. Should the relationship use unique\_ptr, shared\_ptr, or weak\_ptr?

## Your Answer

Correct Smart Pointer Choice

1. Choice

std::unique\_ptr

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Correct Answer  
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## Q5 Replace

10 marks

What should replace delete data in Question 55, assuming data points to an array?

## Your Answer

Replacement for delete data

1. Correct Syntax

delete[] data;

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Q6

Correct Answer  
10 marks

## Q4 Execute

10 marks

```
Which destructors execute? class A { public: ~A() { cout << "A "; } }; class B { public: ~B() { cout << "B "; } }; int main() { try { A a; B b; throw 10; } catch (...) { cout << "Exception"; } }
```

## Your Answer

Executed Destructors Analysis

1. Executed Destructors

Both ~B() and ~A() will execute (in that specific order).



Secure Exit

SIMATS Online Programming Lab

C++ PROGRAMMING

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Correct Answer  
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Q7  
Correct Answer  
10 marks

Q3 Compile

10 marks

Will this compile? If not, identify the exact reason. `class A { public: A(int x) {} }; int main() { A obj; }`

Your Answer

No, this code will not compile.

Exact Reason for Compilation Error

Missing Default Constructor: Class A defines a parameterized constructor `A(int x) {}`. In C++, when any user-defined constructor is declared, the compiler suppresses the automatic generation of the implicit default (no-argument) constructor.

Q2 Virtual

10 marks

What changes if virtual is removed from `Base::show()` in `[class Base { public: virtual void show() { cout << "Base"; } }; class Derived : public Base { public: void show() override { cout << "Derived"; } }; int main() { Base* p = new Derived(); p->show(); }]`

Your Answer

Changes When virtual is Removed from `Base::show()`

1. Compilation Error (override Keyword Violation)

Issue: In `Derived::show()`, the override specifier is explicitly used.

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Q6

Correct Answer  
10 marks

Q1 Find

10 marks

Find the resource-management bug. class Buffer { int\* data; public: Buffer(int size) { data = new int[size]; } ~Buffer() { delete data; } };

Your Answer

Resource-Management Bugs in Buffer Class

1. Mismatched Array Deallocation (delete vs delete[])

Bug: In the constructor, memory is allocated using array new (data = new int[size];), but in the destructor, it is deallocated using single-object delete (delete data;).

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Correct Answer  
10 marks

Q10 Predict

10 marks

Predict the behavior. class A { public: A() {} A(A&&) { cout << "Move"; } }; int main() { A a; A b = std::move(a); }

Your Answer

Behavior Prediction Analysis

1. Output

Move